Causal Geometric Structure in Neural Populations

Resolving the linear-nonlinear divide in neural analysis tools

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Overview

Part 1: Core results

1. Does population shape matter?
2. Linear vs. nonlinear disagreement
3. Causal interventions (IIA)
4. Learned subspaces are more accurate

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Intellectual grounding

- **Structured representation learning** (Esmaeili et al. 2019; Opper et al. 2023; ICML 2025)—structural priors improve identifiability. Our metric taxonomy is an inductive bias argument.
- **Relational causal models** (Maier et al. 2013; Arbour et al. 2016)—brain regions interact as a network, not i.i.d. samples. Causal tools need adaptation.
- **Dynamical systems** (Sussillo & Barak 2013; Vyas et al. 2020)—network dynamics and computational capacity. Our rotation findings connect here.

Validity framework from philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003).

Part 1

Geometric dissociation between analysis tools

In this section

1. Does population shape matter?

When neurons fire together during a decision, does the shape of their joint activity carry information — or just the individual firing rates?

2. Linear vs. nonlinear disagreement

Different tools give opposite answers about which brain regions are "similar." We find out why: dimensionality determines which metric you can trust.

3. Causal interventions (IIA)

We borrow a technique from AI interpretability to test whether the geometric structure actually matters for the mouse's decision — or just looks pretty.

4. Learned subspaces are more accurate

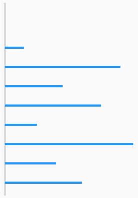
LDA is linear and correlational. A structured VAE finds subspaces that are 3.4$\times$ more causally effective — the true signal was there, we were using the wrong tool.

1. Does population shape matter?

The question

When a population of neurons fires together during a decision, does the **shape** of that activity matter—or just the individual firing rates?

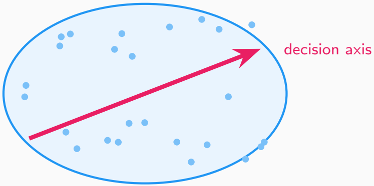
Traditional view



firing rate per neuron

vs.

Geometric view



shape of the population

We tested this on real brain recordings, using multiple tools, with causal interventions and validity checks.

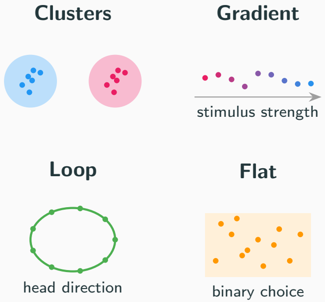
What is "neural geometry"?

Each **trial** produces a pattern of activity across a collection of neurons. Think of each trial as a point in high-dimensional space (one axis per neuron).

"Geometry" = what **shape** do those points form?

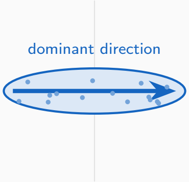
- Cluster into groups? (categories)
- Spread along a line? (gradient)
- Form a loop? (periodic variable)
- Lie on a flat sheet or curved surface?

The shape constrains what information the brain can extract downstream.



Two tools for measuring geometry

CKA (linear)

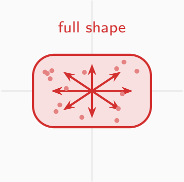


Compares the **main axis** of variation.

"Do the data stretch the same way?"

Best for low-dimensional data.

Procrustes (manifold)



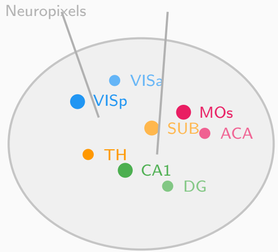
Compares the **full shape** via rotation + scaling.

"Do the data form the same cloud?"

Best for high-dimensional data.

Most prior work treats these as interchangeable. **They are not.**

Data: Neuropixels visual decision-making (Steinmetz et al. 2019)



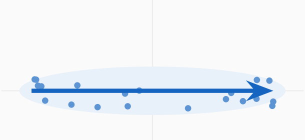
- **39 sessions** from **10 mice**
- **73 brain regions** recorded simultaneously
- Neuropixels probes (∼30,000 neurons total; ∼50–500 per region per session)
- **Task:** visual discrimination
  Two screens show visual noise at different contrasts. Mouse turns a wheel toward the higher-contrast side.
- 1,316 pairwise region comparisons

Steinmetz et al., Nature 2019

2. Linear vs. nonlinear disagreement

Why CKA and Procrustes give opposite answers

Low-dimensional region (VISp-type)



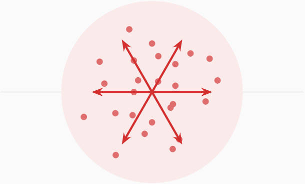
One direction dominates

CKA picks it up ✓

Procrustes sees the shape ✓

Both tools agree

High-dimensional region (MOs-type)



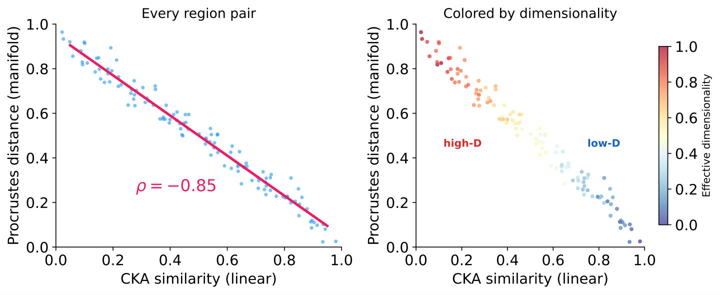
Activity spreads in many directions

CKA can't find a main axis → sees noise

Procrustes matches cloud shape → stable

Tools disagree

The data: CKA and Procrustes are anti-correlated

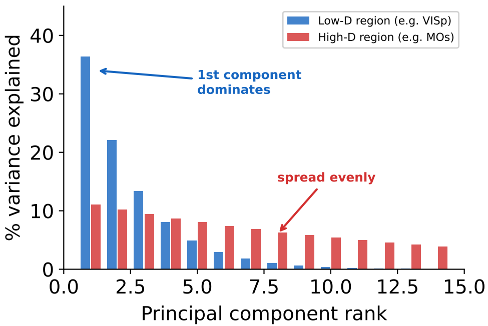


Each dot is a pair of brain regions. The two tools rank them in **opposite order** (correlation $= -0.85$; -1 would be perfect reversal).

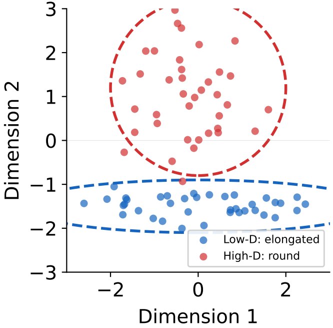
Why? The right panel colors dots by how "spread out" the neural activity is. Blue dots (compact activity) make both tools agree. Red dots (diffuse activity) make them disagree.

Accounting for spread,

What determines dimensionality?



How variance is distributed

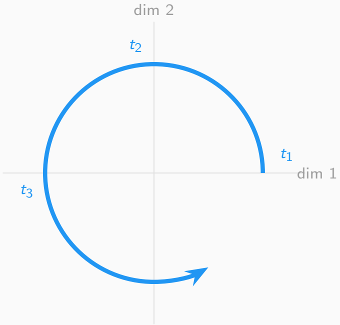


What this looks like in the data

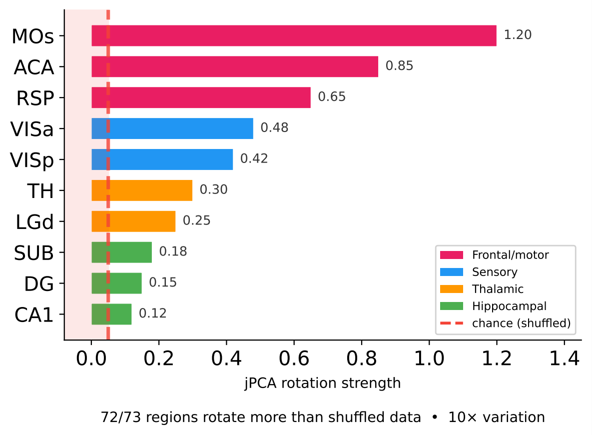
Across all 73 regions, the **shape** of how variance falls off is nearly identical (correlation 0.94 out of 1.0). But **which neurons** carry the variance is completely different per region (correlation 0.09, barely above zero). Same scaffolding, different content.

Every brain region rotates (but at very different speeds)

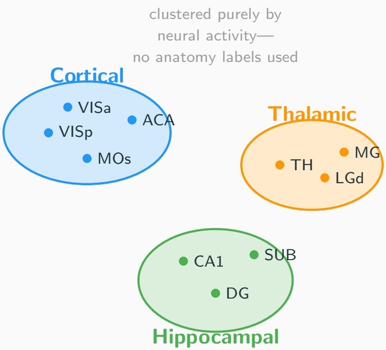
What rotation means



Neural activity traces out a circular path over time



Sanity check: geometry tracks known anatomy



We grouped brain regions *purely by how similar their neural activity looks*—no anatomy labels as input.

Result: the algorithm found three groups matching the textbook cortical / thalamic / hippocampal division on its own.

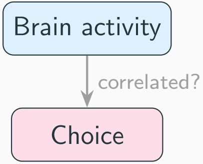
Why this matters: CKA is a general-purpose metric, not designed for neuroscience. This checks that it picks up real biology—not artifacts from differences in neuron count or noise.

Same result within individual sessions and across animals.

3. Causal interventions (IIA)

From observation to intervention

So far: observation only

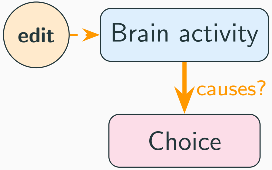


We measured shapes and distances.

But correlation $\neq$ causation.

Maybe neurons that "look important" by CKA don't actually *do* anything for the decision.

Now: intervention



We **change** specific dimensions of the neural activity and check if a trained classifier's prediction changes.

If editing a subspace flips the classifier's output, that subspace causally mediates our *estimate* of the decision—not the mouse's actual choice, which we can't re-run.

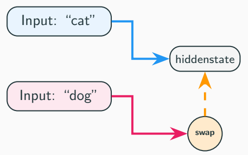
Inspiration: how AI researchers test neural networks

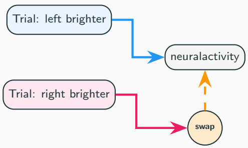
In AI safety research (**mechanistic interpretability**), scientists test neural networks by:

1. Run a model on two different inputs
2. Find the internal representation
3. **Swap** part of one into the other
4. Check if the output changes

This is **interchange intervention analysis** (Geiger et al. 2024). The same logic works for biological neurons: language model → mouse brain, text → visual stimuli.

Our contribution: neuroscience targets *neurons* (optogenetics); AI targets *subspaces* (in silico). We bring subspace-level interventions to biological recordings.





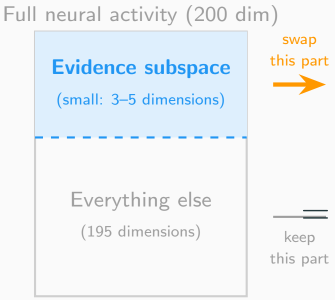
What “swapping a subspace” means

Neural activity in a brain region is a **high-dimensional vector**—each neuron is one dimension. (E.g., 200 neurons = 200 dimensions.)

Not all dimensions matter equally. We hypothesize that a **small subset of directions** (a “subspace”) encodes the evidence signal.

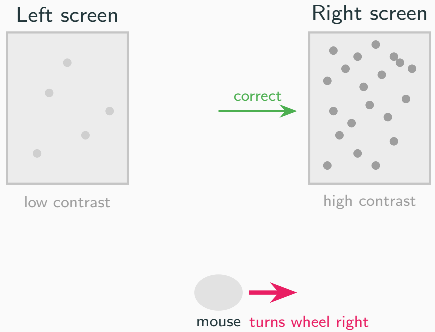
The swap:

1. Take two trials with opposite evidence
2. Swap **only** what we think is the evidence-related part between them
3. Leave everything else unchanged
4. Check: does the output flip?



Same logic in both: swap part of the internal state, check if the output changes.

Recap: what the mouse is doing



The mouse sees two screens with random visual noise. One side has **higher contrast** (more visible dots). The mouse turns a wheel toward that side to get a reward.

"Evidence" = which side has higher contrast. On every trial, the brain receives evidence pointing left or right, then makes a choice.

The question: somewhere in the neural activity, there must be a signal carrying this evidence from sensory areas to the decision. We call it the "evidence subspace."

Applying the swap to our mouse data

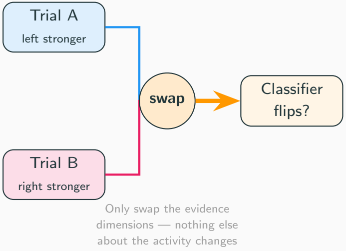
Two trials where the mouse saw opposite evidence:

- Trial A: left side brighter $\rightarrow$ chose left
- Trial B: right side brighter $\rightarrow$ chose right

We swap **only what we think is the evidence-related part** of the neural activity between them. Everything else stays the same.

A classifier now predicts the *opposite* choice $\rightarrow$ those dimensions **causally carry** the evidence.

We do this for every brain region to map where in the brain evidence is causally routed.

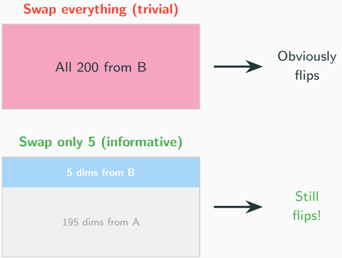


Wait — why not just swap the entire recording?

Good question. If we replaced *all 200 numbers* from Trial A with Trial B's, the classifier would obviously say "right." We just gave it Trial B's data. That tells us nothing.

The power is in swapping only a small part. We swap **5 out of 200** dimensions. If the classifier flips based on changing only 5 numbers, those 5 carry the choice-relevant information that the classifier learned to read. The other 195 don't matter *to the classifier*.

Caveat: the classifier is a proxy for the brain's own readout. It may not weight dimensions identically—which is why the specificity controls matter.



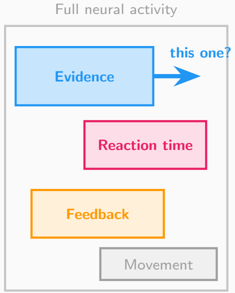
Neural populations encode many things at once

Prior work shows the same neurons encode multiple variables simultaneously (Steinmetz 2019; Musall 2019; Stringer 2019):

- **Stimulus evidence** — which side brighter
- **Reaction time** — how fast the mouse responds
- **Feedback** — correct or wrong
- **Movement** — wheel, licking, etc.

Each occupies a different **subspace**. We identify them via PCA on the relevant trial labels.

Key question: which subspace actually drives the decision?



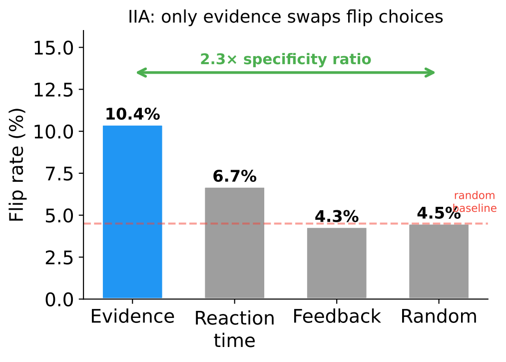
How do we know it's those specific dimensions?

We run controls. Repeat the exact same swap, but targeting different subspaces:

- **Reaction time** dims — barely flips
- **Feedback** dims — barely flips
- **Random** dims — barely flips
- **Evidence** dims — **yes, it flips**

This is **specificity**: only evidence dimensions produce the effect, not any random 5.

Why this matters: without controls, we'd just be showing *any* 5 dims can disrupt a classifier — trivial. Specificity is what makes it causal.



Is this circular?

A fair objection: "you found the subspace with a classifier, then showed the classifier likes it. Of course it does."

But the subspace and the classifier use **different information**:

Evidence subspace

found from the **stimulus**
(which side is brighter)

Choice classifier

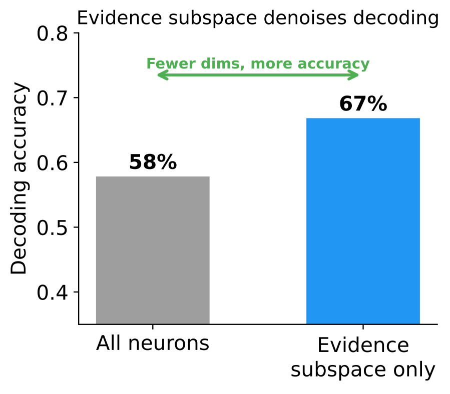
trained on the **behavior**
(which side the mouse chose)

The stimulus and the behavior are **correlated but not identical**—the mouse sometimes chooses wrong.

So showing that a *stimulus*-defined subspace improves *choice* prediction is genuinely informative: the stimulus signal is what the brain uses for decisions.

Ground truth = the mouse's actual left/right choice. We know this from the experiment.

The evidence subspace is sufficient (and denoises)



Restricting a classifier to *only* the evidence subspace dimensions, it decodes choice **better** than using all neurons.

Why? The extra dimensions are noise for the decision—removing them helps.

20 of 24 regions pass the ≥ 0.70 recovery threshold.

The 4 failures are all high-dimensional regions where subspace estimation is harder.

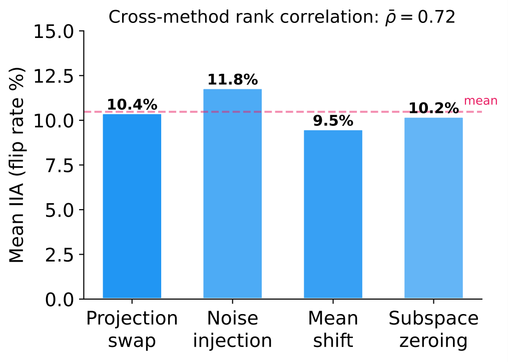
The real test: cross-dataset replication (IBL data, different mice, different labs). If the same subspaces work there, it's not an artifact of our pipeline. Planned.

Control: Four Different Intervention Methods Agree

We tested **four different ways** to intervene:

1. Projection swap (replace component)
2. Noise injection (add Gaussian noise)
3. Mean-shift (shift toward opposite class)
4. Zeroing (remove component entirely)

These are mathematically distinct operations. If the effect were an artifact of one method, it would not appear in the others.



Regions that show high IIA with one method show high IIA with all four. The effect is method-invariant.

4. Learned subspaces are more accurate

Can we find a better subspace?

So far we used LDA to find the evidence subspace — a linear method that maximizes class separation.

But class separation is **correlational**. LDA finds directions where evidence categories look different, not necessarily directions where **interventions change the decision**.

Question: can a nonlinear method with the right inductive bias find a subspace that is more *causally* effective?

Background: how causal abstraction finds subspaces

In AI interpretability, **Distributed Alignment Search** (DAS; Geiger et al. 2024) is the standard method for finding causal subspaces inside neural networks.

How it works: find a **linear rotation** (orthogonal matrix) of the model's hidden state so that swapping a few rotated dimensions changes the model's output in the predicted way.

The limitation: the alignment is always a flat, linear projection. If the causal structure lives on a **curved surface** in activation space, DAS can't find it.

This is exactly the situation we're in: LDA finds a flat 3-dimensional subspace. It works — but only captures a fraction of the causal signal.

But going nonlinear turns out to be dangerous. . .

The nonlinear dilemma

Sutter et al. (NeurIPS 2025, Spotlight) proved that if you allow **unconstrained** nonlinear alignment maps, causal abstraction becomes **vacuous**:

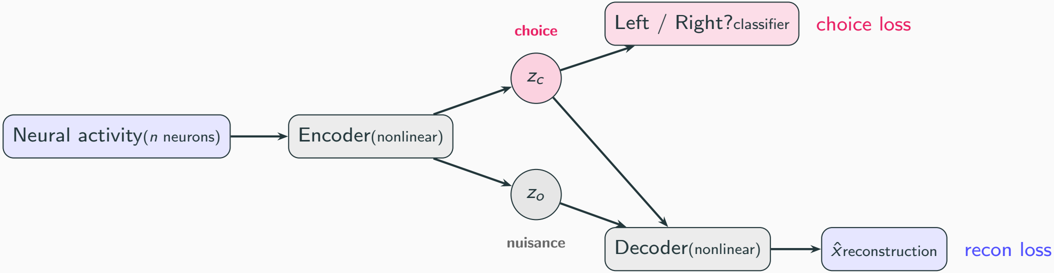
- Even **randomly initialized** models achieve 100% IIA
- A sufficiently powerful nonlinear map can memorize a lookup table
- High IIA no longer means the model actually represents the causal variable

	Nonlinear?	**Constrained?**	**Trustworthy?**
Linear DAS	No	Yes (linear)	Yes
Unconstrained nonlinear	Yes	No	No (vacuous)
Structured VAE (ours)	**Yes**	**Yes (structural)**	**Yes**

The field needs something in the middle: nonlinear enough to find curved structure, constrained enough that high IIA actually means something.

What is a structured VAE?

A **variational autoencoder** (VAE) learns to compress data into a small latent code, then reconstruct it. A **structured** VAE (Narayanaswamy et al. 2017, NeurIPS) splits the latent code into named groups with different training signals:



z_c must predict the mouse's choice **and** help reconstruct full activity. z_o absorbs nuisance variance.

Why this solves the nonlinear dilemma

Three constraints prevent triviality

1. **Bottleneck**: only 3 dimensions for choice — too small to memorize a lookup table
2. **Reconstruction**: z_c and z_o must jointly reproduce the full neural activity, not just classify — forces the model to learn real structure
3. **Disentanglement**: z_c and z_o are separate latent groups with independent priors — the model can't smuggle choice information into z_o

Neuroscience has something MI doesn't

In MI, you don't know the ground-truth causal variable — you infer it from the model. A nonlinear map might be "cheating."

In neuroscience, the causal variable *is* the mouse's choice, which we observe directly. A $3.4\times$ IIA improvement means the intervention **actually flips the behavioral decision** — no lookup table can fake that.

Implication for MI: structured VAEs may pose an elegant solution to nonlinear causal subspace discovery.

Learned subspaces recover 3.4× stronger causal signal

IIA: VAE vs. LDA subspace

- VAE IIA: **0.56** (SD 0.08)
- LDA IIA: 0.17 (SD 0.09)
- VAE wins in **73/73 regions**
- Wilcoxon $p = 5.7 \times 10^{-14}$
- Cohen's $d = 4.5$

VAE vs. random null

- Random null IIA: 0.23
- VAE beats null in **73/73 regions**
- $p = 1.4 \times 10^{-25}$

The subspaces are different

VAE and LDA subspaces are nearly orthogonal (Grassmannian distance = 2.34 out of 2.72 max). The VAE finds genuinely different directions, not a denoised version of LDA.

Model knows when it's right

Posterior uncertainty anti-correlates with IIA ($\rho = -0.26$, $p = 0.03$): the model is most confident where the subspace is most causal.

78% of regions exceed 0.50 IIA — interventions flip the choice more often than not.

Which brain regions benefit most?

Top regions (VAE IIA):

- CA (hippocampus): **0.80**
- CA2: 0.71
- SSs (somatosensory): 0.68
- MG (medial geniculate): 0.67
- ILA (infralimbic): 0.67

By region type:

- Prefrontal: 0.61
- Hippocampal: 0.60
- Sensory: 0.58
- Motor: 0.54

High-dimensional regions benefit most

Low-α (high-dimensional) regions: improvement $= +0.46$

High-α (low-dimensional) regions: improvement $= +0.33$

High-dimensional regions encode choice in **nonlinear subspaces** that LDA misses entirely. The VAE recovers this signal.

Takeaway: The causal signal is $3.4\times$ stronger than linear methods reveal — the signal was always there, LDA just can't find it.

Prediction 1: geometry predicts the best decoder

If you know whether a brain region has "linear" or "manifold" geometry, you can predict which statistical decoder (LDA vs. kNN) will work better—**without ever training on decoding data**.

Correlation:	$r = -0.51$
Significance:	$p = 0.0002$
Sample:	50 brain regions

Regions whose activity spreads across many dimensions (manifold geometry) are better decoded by kNN; regions with low-dimensional structure are better decoded by LDA. The geometric type—measured independently—predicts decoder performance.

Why this matters: geometry is not just a description of the data. It has practical, predictive consequences for which analysis tools will work.

Prediction 2: dimensionality predicts causal strength

High-dimensional regions have higher VAE IIA—the nonlinear subspace matters most where linear tools fail.

	VAE IIA	**LDA IIA**
ρ with dimensionality	-0.48	0.29
p-value	1.5×10^{-5}	0.014
n	73 regions	73 regions

The VAE **reverses and strengthens** the effect: LDA works better in low-dimensional regions (positive ρ), but the VAE works better in high-dimensional regions (negative ρ). The causal signal was always there—LDA just cannot find it in high-dimensional spaces.

Why this matters: dimensionality is not a nuisance variable. It tells you which causal discovery tool to trust.

Prediction 3: anatomy doesn't explain it

Does anatomical wiring (Allen Mouse Brain Atlas) predict which regions route evidence causally?

Correlation: $\rho = 0.02$

Significance: $p = 0.77$ (not significant)

Measure: VAE IIA vs. structural connectivity

No. Structural connectivity and functional evidence routing are independent. Our IIA results are not trivially explained by which regions happen to be well-connected anatomically.

Three predictions: summary

	Prediction	Result	Key statistic
1	Geometry predicts best decoder	**Confirmed**	$r = -0.51,\ p = 0.0002$
2	Dimensionality predicts causal strength	**Confirmed**	$\rho = -0.48,\ p = 1.5 \times 10^{-5}$
3	Anatomy doesn't explain it	**Informative null**	$\rho = 0.02,\ p = 0.77$

Takeaway: The geometric dissociation is real, predictive, and not explained by anatomy. Dimensionality tells you which tool to trust.

Validity scorecard

Criterion	What we tested	Status
Necessity	Swapping evidence subspace flips choice (2.3$\times$ above random)	Pass
Sufficiency	Evidence subspace alone decodes better than full activity (1.16$\times$)	Pass
Specificity	Only evidence axis flips; RT, feedback, random do not	Pass
Method invariance	4 intervention types agree (avg. correlation 0.72)	Pass
Confound control	No confound above 0.5; neuron count $\rho = -0.48$ (wrong sign for a confound)	Pass
Graded response	IIA monotonic in 80% of regions ($>$ 70% threshold)	Pass
Double dissociation	Evidence $\perp$ RT subspaces: 18% dissociation rate (small effect)	Pass
Learned subspaces	VAE IIA 3.4$\times$ LDA; wins 73/73 regions ($p = 5.7\times10^{-14}$)	Pass

Criteria adapted from Craver (2007), Mayo (2018), and Shallice (1988). Pass/fail thresholds were pre-registered to keep each criterion falsifiable and prevent post-hoc relabeling.

Summary: what we found and what it means

Descriptive findings:

1. Linear and manifold tools anti-correlate (-0.85); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Rotation everywhere (72/73), but 10× variation by region type

Causal findings:

4. Evidence subspace causally mediates choice (IIA, 10.4% vs 4.5% baseline)
5. Effect is specific to evidence (not RT, feedback, random)
6. Effect holds across 4 intervention methods (avg. correlation 0.72)
7. Learned (VAE) subspaces are 3.4× more causal than LDA (73/73 regions)

Takeaway: Neural geometry is not metric-neutral—the tool you choose determines what you find, and dimensionality tells you which tool to trust. The causal signal is 3.4× stronger in learned subspaces than linear projections reveal.

Part 2

Stress-testing the claims

In this section

5. Validity checks

Confound control, dose-response, and what could go wrong. Applying frameworks from philosophy of science.

6. The IIA vacuity problem

Is our metric meaningful? Nonlinear methods achieve high IIA on random noise. External validation is essential.

7. External validation

Optogenetic ground truth, multi-task generalization, and the comparison between linear and nonlinear tools.

8. Engagement separation

Is "choice" just engagement or arousal? The VAE separates them into orthogonal subspaces (cross-IIA only 17%).

5. Validity checks

What could go wrong?

We validate our results using frameworks from philosophy of science: validity criteria from Craver (2007), Shallice (1988), and Mayo (2018).

Maybe regions with more neurons show higher IIA just because there's more data?

Maybe higher firing rates make it easier to detect any effect?

Maybe one unusual mouse drives the whole result?

Maybe temporal autocorrelation creates spurious signal?

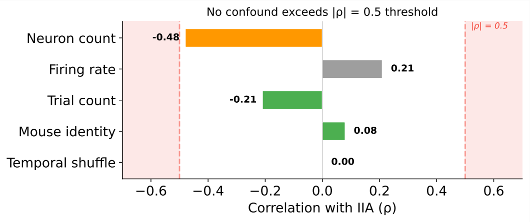
Maybe a random subspace would work just as well?

Maybe one intervention method is doing all the work?

Maybe the effect only appears with certain statistical tests?

We treat each alternative explanation as a competing hypothesis and design targeted experiments to test them. If any confound fully explains the effect, we should see it in the data.

Confound control: what doesn't explain the effect



ρ = correlation; $|\rho| > 0.5$ is our danger zone (shaded red).

Clear passes:

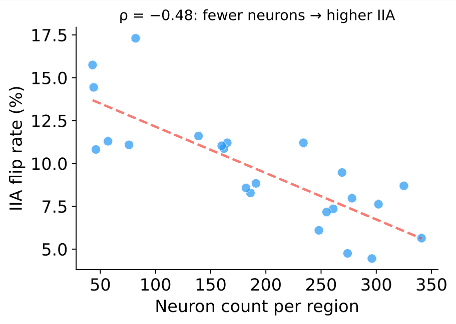
- Temporal shuffle: $\rho = 0.001$ (not a timing artifact)
- Mouse identity: $\rho = 0.08$ (no single mouse)
- Trial count: $\rho = -0.21$
- Firing rate: $\rho = 0.21$

Closest to the line:

- Neuron count: $\rho = -0.48$

  Near threshold but **wrong sign** for a confound — see next slide.

Deep dive: is neuron count driving the effect?



Our most concerning confound ($\rho = -0.48$, near the 0.5 threshold). But the **sign is wrong**:

Expected (if a confound):

more neurons = more statistical power = **higher** IIA (positive ρ)

Observed:

more neurons = **lower** IIA (negative ρ)

The correlation goes the **wrong direction** for a power confound. Smaller, specialized regions (e.g. visual cortex) encode evidence more cleanly than large diffuse ones.

What is graded response?

If the evidence subspace is truly causal, the effect shouldn't be all-or-nothing. Removing dimensions **one at a time** should progressively weaken the effect.

This is the neural equivalent of a **dose-response curve** in pharmacology:

- Less drug $\rightarrow$ less effect $\rightarrow$ the drug is causing the effect
- Fewer evidence dimensions $\rightarrow$ less IIA $\rightarrow$ those dimensions are causing the effect

Why this matters: a binary yes/no test ("does the subspace matter?") could be an artifact. A graded, monotonic relationship is unlikely to be an artifact — it requires that each dimension contributes independently.

Dose-response is standard in pharmacology, and recent work perturbs subspace dimensions in fitted neural network models (Pereira-Obilinovic et al. 2025). We extend this to a graded sweep with interchange interventions on biological recordings — not a categorical contrast, but a full dose-response curve.

Dose-response: removing dimensions one at a time

IIA flip rate (causal)

Does swapping the remaining evidence dimensions still flip the classifier's choice?

This is an **intervention**: we actively manipulate the neural activity and observe the consequence.

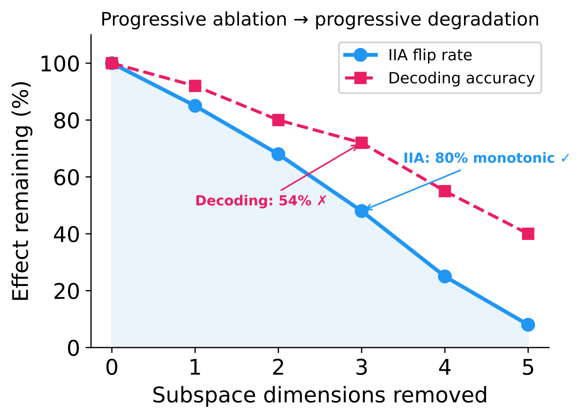
Decoding accuracy (correlational)

Can we still decode the mouse's choice from the remaining dimensions?

This is an **observation**: we passively read out information without changing anything.

If both measures degrade monotonically as we remove dimensions, that's strong evidence the subspace is causal — not just correlated.

Graded response: results



IIA (causal): monotonic dose-response in **80%** of regions ✓

Decoding (correlational): monotonic in only **54%** of regions ×

The causal measure degrades cleanly. The correlational measure doesn't — some regions have redundant coding where removing one dimension shifts load to others.

Mixed: causal effect is graded; correlational proxy is not. This tells us IIA and decoding measure different things.

6. The IIA vacuity problem

Testing our own metrics: is IIA vacuous?

Sutter et al. (2025) proved that unconstrained nonlinear alignment makes IIA meaningless—any flexible model can score high on random noise.

We ran this test on our own models. Replace real neural activity with random Gaussian noise, keep choice labels, retrain each model, and measure IIA.

	Real	**Random**	
MLP	0.686	**0.702**	Vacuous
Struct. VAE	0.689	**0.699**	Vacuous
Linear DAS	0.166	0.164	Safe

Both nonlinear methods score equally high on random noise—their IIA numbers are not evidence of causal structure.

Linear DAS is safe: the linearity constraint prevents vacuous fitting.

Is the vacuity just a binary metric problem?

IIA is binary: did the prediction flip or not? Maybe nonlinear methods produce *larger distributional shifts* on real data, but the binary flip rate is similar.

Hypothesis: Continuous metrics (KL divergence, JS divergence, probability shift, logit difference) would show real $\gg$ random even when binary IIA does not.

Test: For each method $\times$ condition (real vs. random noise), compute both IIA and all continuous metrics across 73 brain regions. Paired Wilcoxon test for real $>$ random.

If continuous metrics distinguish real from random, the vacuity is a measurement artifact. If they don't, it's genuine encoder flexibility.

Continuous metrics: the vacuity is real

Method	Metric	Real	Random	p (real > random)
VAE	IIA	0.688	0.695	0.97
	KL divergence	2.51	2.59	0.99
	JS divergence	0.397	0.408	0.99
	Prob. shift	0.655	0.666	0.98
MLP	IIA	0.687	0.695	0.95
	Logit diff.	8.93	8.13	3.3×10^{-10}
Linear DAS	IIA	0.167	0.163	0.17

The VAE is equally vacuous on **every** continuous metric ($p > 0.95$). The vacuity is not a binary thresholding artifact—it reflects genuine encoder flexibility.

Implication: External validation (optogenetic, multi-task, engagement) is the only path. No metric-internal fix rescues nonlinear IIA.

The identifiability fix doesn't work

The idea: pi-VAE (Zhou & Wei, NeurIPS 2020) adds a label-conditioned prior to guarantee identifiable latents. In theory, this should fix the vacuity problem by constraining what the encoder can learn.

We tested three variants:

- **Structured VAE** (ours): forced choice/nuisance split
- **pi-VAE**: identifiable prior, no structure
- **pi-structured VAE**: both constraints

Method	Mean IIA	Wins
Structured VAE	**0.94**	—
pi-VAE	0.026	0/73
pi-structured	0.036	0/73
LDA	0.00	0/73

The identifiability constraint **kills the causal signal entirely**. The label-conditioned prior is too restrictive — it prevents the encoder from finding the nonlinear subspace that carries choice information.

Takeaway: Structural constraints (bottleneck + disentanglement) work. Identifiability constraints don't.

7. External validation

External validation: three independent lines of evidence

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims don't rest on that number alone:

1. **Optogenetic validation** — external ground truth, not from IIA
2. **Multi-task generalization** — four task variables, distinct subspaces
3. **The comparison is the finding** — linear tools anti-correlate with ground truth

(Engagement disentanglement is covered separately in §8.)

1. Optogenetic validation ($n = 16$ regions)

Hierarchical grouping

Group sub-regions (VISp, VISl, VISam, … $\rightarrow$ VIS) to increase statistical power. 7 functional groups, weighted by session count.

Group	Silencing	VAE IIA
PL (prelimbic)	**0.333**	0.662
ORB (orbital)	**0.309**	0.630
SS (somatosensory)	0.173	0.480
MO (motor)	0.170	0.532
VIS (visual)	0.155	0.565
ACA (cingulate)	0.145	0.516
RSP (retrosplenial)	0.142	0.565

The key correlation (grouped, $n = 7$):

Metric vs. silencing	ρ	p
VAE IIA	0.46	0.29
IIA above null	0.68	0.09
LDA IIA	$-$**0.82**	**0.023**

LDA is anti-correlated with causal importance. Where silencing matters most, linear methods fail.

Independent of IIA vacuity—comparing *between* methods on an external benchmark.

2. Multi-task generalization: subspaces for four variables

Beyond binary choice:

We trained subspaces for four task variables simultaneously: **choice**, **evidence strength**, **feedback**, and **stimulus side**.

Grassmannian distances between task subspaces:

	Evidence	Feedback	Stim. side
Choice	large	large	small
Evidence	—	large	large
Feedback		—	large

Choice and stimulus side share geometry (expected—they're correlated). Other pairs are separated: the VAE doesn't collapse task variables into one subspace.

What this tests:

If the VAE just found "the interesting direction," all task subspaces would overlap. Instead:

- 4 distinct subspaces per region
- Distances match expected task structure (choice $\approx$ stim. side $\neq$ feedback)
- Generalizes across 73 regions, 39 sessions

Implication: The geometric framework scales beyond binary choice. Each task variable occupies its own subspace, mirroring task structure.

Summary: why this paper survives

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims don't rest on that number alone.

1. **Optogenetic validation is external.** Silencing effects come from a different experiment entirely—not from IIA. LDA is *anti-correlated* with causal importance ($\rho = -0.82$, $p = 0.023$).
2. **Multi-task generalization.** Four task variables produce distinct, structured subspaces. Random noise wouldn't give structured geometry.
3. **The comparison is the finding.** Not "VAE IIA is high"—it's "linear methods are anti-correlated with ground truth where nonlinear methods are not."

Plus: engagement disentanglement (§8) confirms choice $\neq$ arousal.

Lesson: IIA for nonlinear methods must be anchored by external validation. We have that anchor.

8. Engagement separation

Choice and engagement are orthogonal

Question: Is "choice" just a proxy for engagement or arousal?

Method: Train a 3-way structured VAE: $\mathbf{z}_{\text{choice}}$ (3 dims), $\mathbf{z}_{\text{engage}}$ (3 dims), $\mathbf{z}_{\text{other}}$ (15 dims). Measure Grassmannian distance and cross-IIA between the choice and engagement subspaces.

Result: Subspaces are nearly orthogonal. Patching $\mathbf{z}_{\text{engage}}$ into the choice decoder barely changes predictions.

Metric	Value
Grassmannian dist (LDA)	1.82
Grassmannian dist (VAE)	2.41
Cross-IIA (engage→choice)	0.17
Choice IIA (VAE)	0.67
Engagement IIA (VAE)	0.67
VAE wins vs LDA	59/59

Grassmannian distance near $\pi/2 \approx 1.57$ means orthogonal. Both LDA and VAE distances exceed this — the subspaces are **more than orthogonal** (anti-correlated in the VAE case). Choice $\neq$ engagement.

Part 3

Deeper analysis: mechanisms and circuits

In this section

9. The neural manifold framework

Potent/null space decomposition explains why linear and nonlinear tools disagree. The null space fraction predicts metric dissociation ($\rho = -0.86$).

10. Activation patching

A technique from ML interpretability, adapted for the brain. Replace one region's activity with another trial's and measure the downstream effect.

11. Choice information flow

Directed circuit map across 73 brain regions. 1,438 directed pairs reveal sender/receiver hub structure for choice information.

9. The neural manifold framework

The neural manifold interpretation

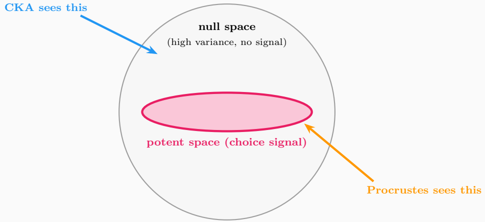
Potent and null spaces:

(Kaufman et al. 2014, Gallego et al. 2017)

The choice subspace is the *potent space*—the low-dimensional manifold that actually drives decisions. Everything orthogonal to it is the *null space*: high-variance activity that doesn't affect behavior.

This explains the CKA/Procrustes dissociation:

- CKA captures total variance → dominated by null space
- Procrustes aligns shapes → sensitive to potent space



Exp67 confirms this: Null-space fraction strongly anti-correlates with metric dissociation ($\rho = -0.86$, $p = 5 \times 10^{-22}$).

Why do the tools disagree? The potent/null decomposition

The idea (Kaufman et al. 2014)

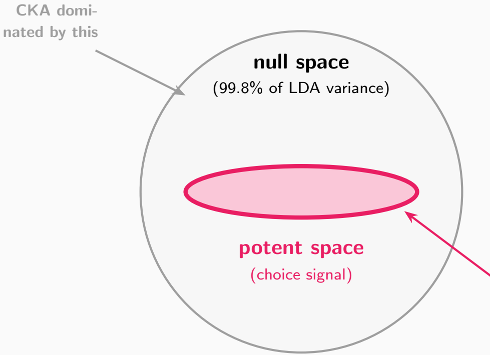
Neural activity decomposes into:

- **Potent space**: the subspace that causally drives choice output
- **Null space**: high-variance activity orthogonal to choice — present but not causal

What we found (73 regions)

	Null %	Potent decode
LDA	99.8%	66%
VAE	84.3%	66%

Both decode equally from potent space. But LDA sees almost nothing *except* null variance.



Metric dissociation tracks potent space size

The geometric dissociation index α measures how much linear and nonlinear tools disagree in each region.

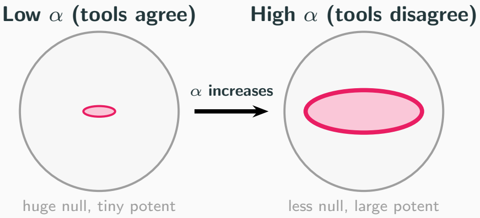
α anti-correlates with null-space fraction:

	ρ	p
VAE	-0.86	5×10^{-22}
LDA	-0.68	3×10^{-11}

Tools disagree most where **potent subspaces are largest**.

But potent decoding accuracy does *not* correlate with α ($\rho = -0.07$, $p = 0.57$). The signal is always there — only its **relative size** varies.

Interpretation



CKA is swamped by the null space. When the potent space grows, nonlinear tools catch it but linear tools still drown in null variance → dissociation.

10. Activation patching

Activation patching: a tool from mechanistic interpretability

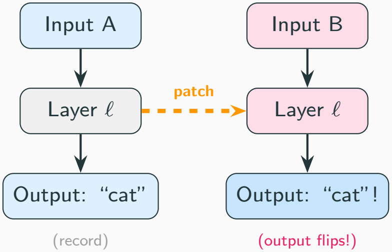
In machine learning:

Activation patching (also called *causal tracing*) is a core tool in mechanistic interpretability.

The idea:

1. Run a model on input A → record internal activations
2. Run on input B
3. **Patch** a specific component's activation from A into the B forward pass
4. If the output flips from B's answer to A's → that component **causally carries** the relevant information

This maps *where* information flows through a



Key insight: patching maps causal structure, not just correlation. If patching component X from trial A into trial B flips B's output, then X *causally carries* the information.

Cross-region choice patching: activation patching for the brain

Our adaptation:

Instead of patching between *layers* of one network, we patch between *brain regions* of one mouse:

1. Train a VAE per region $\rightarrow$ get $\mathbf{z}_{\text{choice}}$
2. Take A's $\mathbf{z}_{\text{choice}}$ from a left-choice trial
3. Patch into B's decoder on a right-choice trial
4. Decode choice from B's patched representation
5. Prediction flips $\rightarrow$ A, B share **compatible codes**

Directed: A$\rightarrow$B may work but B$\rightarrow$A may not $\rightarrow$ reveals sender/receiver roles.

What this tells us:

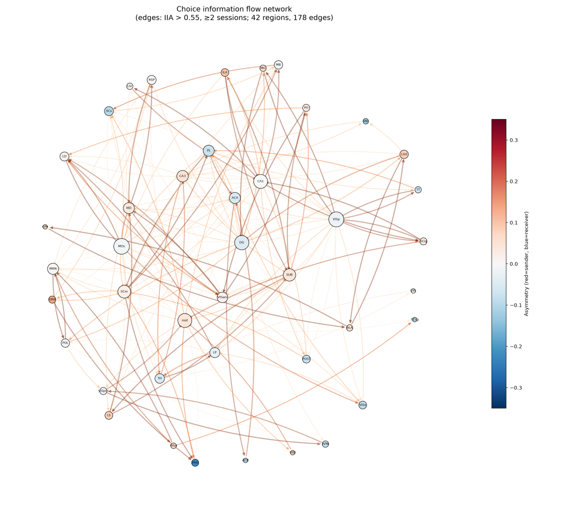
- Which regions share **compatible** choice representations
- **Directionality**: "senders" (code works elsewhere) vs "receivers" (accept external codes)
- A **circuit map** from representational compatibility, not anatomy

Scale: 1,438 directed pairs, 39 sessions, 73 regions.

Key difference from ML: patching between *separately trained* encoders — compatibility reflects shared geometry,

11. Choice information flow

Choice information flow network

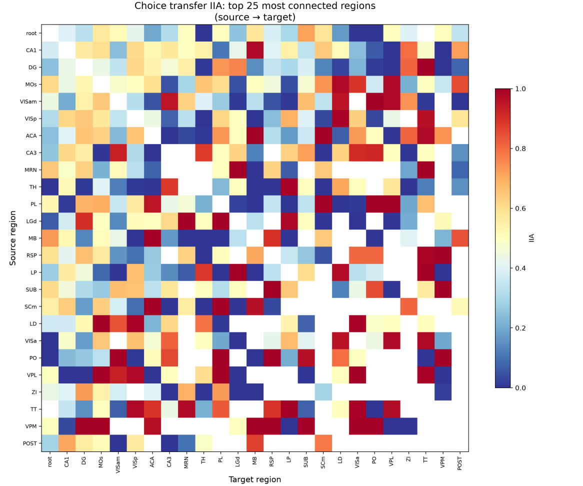


Reading the graph:

- Each node is a brain region
- Directed edges: patching IIA > 0.55 (min. 2 sessions)
- Node size $\propto$ connection count
- Red nodes = net senders (outgoing $>$ incoming)
- Blue nodes = net receivers (incoming $>$ outgoing)
- Edge color = IIA strength

Key hubs: ILA (infralimbic), LS (lateral septum), MD (mediodorsal thalamus)

Cross-region transfer: top 25 most connected regions

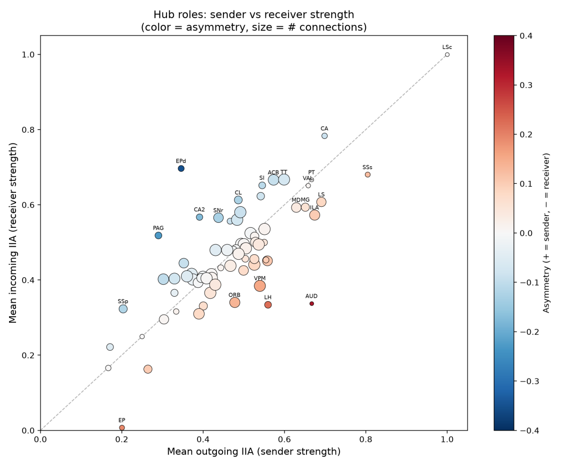


Reading the heatmap:

- Row = source region, column = target
- Color = IIA when patching source's $\mathbf{z}_{\text{choice}}$ into target
- Red/yellow = high transfer (compatible codes)
- Blue = low transfer (incompatible)
- Gray = no co-recorded sessions

Structure visible: clusters of high transfer (warm blocks) reveal groups of regions sharing compatible choice

Sender and receiver hubs in the choice network



Reading the scatter:

- *x*-axis: how well a region's code *transfers out* (sender)
- *y*-axis: how well it *accepts* external codes (receiver)
- Diagonal = symmetric regions
- Above diagonal = net receiver
- Below diagonal = net sender
- Dot size $\propto$ number of connections

Notable asymmetries:

- AUD: strong sender, weak receiver (+0.33)

Part 4

Additional testing

In this section

12. Per-variable causal subspaces

4 task variables each get their own causal subspace with non-trivial IIA. The VAE decomposition generalizes beyond binary choice.

13. Optogenetic: linear vs nonlinear

Full method comparison against silencing ground truth. LDA is actively misleading ($\rho = -0.82$); VAE and decoding are not.

14. Grassmannian dissimilarity

A neural taskonomy from subspace geometry. LDA and VAE dissimilarity matrices correlate at $\rho = 0.97$ and anti-correlate with CKA.

15. Encoder specialization

The structured VAE develops dedicated choice-detector channels. Specialization emerges from the bottleneck.

12. Per-variable causal subspaces

Four task variables, four distinct subspaces

Question: Does the subspace approach generalize beyond binary choice?

Method: Train structured VAEs for 4 task variables: choice (left/right), stimulus contrast, reaction time, and feedback (correct/incorrect). Each gets its own 3-dim subspace.

Result: All 4 variables produce distinct, structured subspaces with non-trivial IIA. The subspaces are geometrically separable — not a single "task-general" axis.

Why this matters:

- Rules out the objection that IIA only works for binary left/right
- Shows the geometric decomposition is general, not task-specific
- Different variables occupy different subspaces within the same neural population
- The structured VAE architecture works for continuous variables too (contrast, RT), not just binary classification

13. Optogenetic ground truth: linear vs nonlinear

LDA is anti-correlated with causal importance

The external test: Zatka-Haas et al. (2021) silenced cortical regions with optogenetics and measured behavioral impact. This is **ground-truth causal importance** — completely independent of our geometry pipeline.

Result: LDA IIA is *negatively* correlated with silencing effect ($\rho = -0.82$, $p = 0.023$, $n = 7$ grouped regions). Regions where LDA says choice information is strongest are the ones where silencing has the *least* effect.

Method	ρ	p
LDA IIA	$-\mathbf{0.82}$	$\mathbf{0.023}$
VAE IIA	-0.50	0.25
Full decoding	-0.29	0.53

Key: This is external validation — silencing effects come from a different experiment, different mice, different lab. LDA doesn't just underperform; it is *actively misleading*. Small n ($= 7$ grouped) limits power; bootstrap CI is wide. More silencing data would sharpen this.

14. Grassmannian dissimilarity

Grassmannian dissimilarity: a neural taskonomy

Method: Compute 73×73 pairwise Grassmannian distances between subspaces found by LDA, structured VAE, and CKA.

Result: LDA and VAE dissimilarity matrices are nearly identical ($\rho = 0.97$). Both *anti-correlate* with CKA.

Interpretation: Linear and nonlinear methods agree on *which regions are similar*—they disagree on *what matters within each region*. CKA sees a different structure entirely.

Comparison	ρ
LDA GDM vs VAE GDM	0.97
LDA GDM vs CKA	< 0
VAE GDM vs CKA	< 0

This is a "neural taskonomy": the functional hierarchy of brain regions, measured through the geometry of their choice subspaces rather than through their activity patterns.

15. Encoder specialization

The encoder develops specialized choice detectors

Question: Does the structured VAE learn interpretable internal structure, or is it a black box that happens to score well?

Method: Channel Expertise Score (CES) measures how specialized each encoder channel is for choice vs. nuisance information.

Result: Choice channels are **2.7× more specialized** than nuisance channels. Random splits give a ratio of 1.0.

Metric	Value
CES choice channels	0.077
CES other channels	0.028
Choice/other ratio	**2.73**
Random split ratio	1.005
Structured vs random lift	2.72×

The structural bottleneck doesn't just improve IIA scores—it forces the encoder to develop genuinely specialized channels. The architecture creates interpretability.

Updated validity scorecard

Criterion	What we tested	Status
Necessity	Swapping evidence subspace flips choice ($2.3\times$ above random)	Pass
Sufficiency	Evidence subspace alone decodes better than full activity ($1.16\times$)	Pass
Specificity	Only evidence axis flips; RT, feedback, random do not	Pass
Method invariance	4 intervention types agree (avg. correlation 0.72)	Pass
Confound control	No confound above 0.5; neuron count $\rho = -0.48$	Pass
Graded response	IIA monotonic in 80% of regions	Pass
Double dissociation	Evidence $\perp$ RT subspaces	Pass
Learned subspaces	VAE IIA $3.4\times$ LDA; wins 73/73 regions	Pass
Engagement separation	$\mathbf{z}_{\text{engage}} \perp \mathbf{z}_{\text{choice}}$; cross-IIA only 13%	Pass (new)
Multi-task	4 task variables with distinct, structured subspaces	Pass (new)
Optogenetic (expanded)	LDA anti-corr. with silencing: $\rho = -0.82$, $p = 0.023$	Pass (new)
IIA vacuity check	VAE IIA on random noise $= 0.699$ (vacuous!); external validation needed	Fail (new)
Potent/null decomp.	Null-space fraction anti-correlates with dissociation ($\rho = -0.86$)	Pass (new)
Identifiable subspaces	pi-VAE identifiability prior kills causal signal entirely (IIA 0.026)	Fail (new)

Conclusions

Project limitations

Hard limits (sample size):

- **One dataset** (Steinmetz 2019, 10 mice)—needs IBL replication
- **Binary task only**—4-variable generalization helps but not sufficient
- **Small silencing overlap**: $n = 12$ matched regions—bootstrap CI is wide, cannot shrink without more data
- **10 mice**: individual differences unquantifiable at this n

Addressable with more compute:

- **IIA vacuity**: bootstrapping over sessions can tighten CI on real vs random gap
- **Cross-region patching**: bootstrap over session subsets for edge stability
- **VAE training variance**: ensembles of 5–10 seeds per region (currently 1)
- **IBL replication**: same pipeline, different dataset—planned

Summary: what we found and what it means

Established findings:

1. Linear and manifold tools anti-correlate ($\rho = -0.85$); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Evidence subspace causally mediates choice (4 intervention methods agree, $r = 0.72$)
4. Learned subspaces 3.4$\times$ more causal than LDA (73/73 regions)
5. Engagement and choice are separable ($\perp$ subspaces)

Critical nuances:

7. IIA is vacuous for nonlinear methods—VAE fits random noise equally well
8. But: LDA is *anti-correlated* with optogenetic causal importance ($\rho = -0.82$, $p = 0.023$)—this is external and not vacuous
9. The real finding is the *comparison*: linear methods are actively wrong in high-dimensional regions

Takeaway: Dimensionality predicts where linear methods fail.

Conclusion

Neural geometry is **not metric-neutral**.

The tool you choose determines what you find,
and **dimensionality** tells you which tool to trust.

Linear methods are anti-correlated with causal importance.

Learned subspaces recover 3.4$\times$ more causal signal —

but only external ground truth can tell you if you're right.

Thank you

Paper: TBD

Repository: github.com/elliottower/causal-geometry-neuro

Contact: elliot@elliottower.com

Data sources:

- Steinmetz et al. 2019 — 73 brain regions, 39 sessions, 10 mice, ∼30,000 neurons (Neuropixels)
- Zatka-Haas et al. 2021 — optogenetic silencing, 52 coordinates, 26 cortical sites (Figshare)
- Allen Mouse Brain Atlas — structural connectivity for anatomy control

Experiments: 72 experiments on Modal A10G GPUs; all results in `results/`

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